



ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA
CAMPUS DI FORLÌ

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Natural Language Processing

Lesson 4. Rule-based Sentiment Analysis (+ Naïve Bayes')

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Previously

- Pre-processing (e.g., tokenisation, stemming, stopwording)
- (Binary) BoW representation
- Dot product

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Sentiment Analysis (with VADER)

Sentiment Analysis

It **does not** refer to actual sentiment (e.g., love or hate)¹
It is about **positive** and **negative** perceptions (plus **neutral**)



This monitor is definitely a good value. Does it have superb color and contrast? No. Does it boast the best refresh rate on the market? No. But if you're tight on money, this thing looks and preforms great for the money. It has a Matte screen which does a great job at eliminating glare. The chassis it's enclosed within is absolutely stunning.

POSITIVE



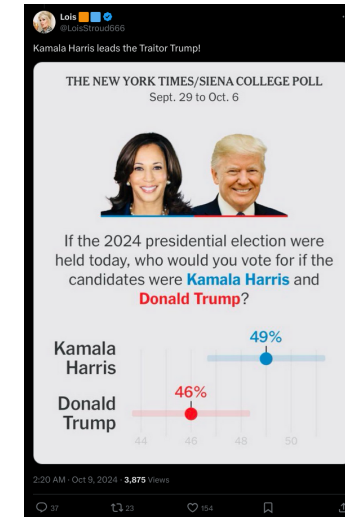
His [ssa] didnt concede until July 12, 2016. Because he was throwing a tantrum. I can't say this enough: [kcuF] Bernie Sanders.

NEGATIVE

From (Lane et al., 2019, p. 62–65)

¹That's emotion analysis; e.g., Fernicola et al. (2020); Zhang et al. (2022)

Sentiment Analysis



<https://x.com/LoisStroud666/status/1843808652802801745>

Valence Aware Dictionary for sEntiment Reasoning
(Hutto and Gilbert, 2014)²

- It features a lexicon where tokens have associated and their "sentiment" score
- It counts all tokens belonging to each category: [pos, neu, neg] ... and combine them to determine the sentiment

</> Let us see it working

²<https://ojs.aaai.org/index.php/ICWSM/article/view/14550>
<https://github.com/cjhutto/vaderSentiment>

Into Machine Learning

Machine Learning

“[...] an umbrella term for **solving problems** for which development of algorithms by human programmers would be cost-prohibitive”

“[...] the problems are solved by helping machines **“discover” their “own” algorithms**, without needing to be explicitly told what to do by any human-developed algorithms.”

https://en.wikipedia.org/wiki/Machine_learning

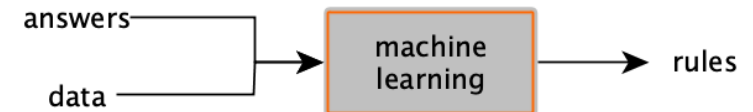
Machine Learning

A change of paradigm

From hand-crafted rules



To training



Diagrams borrowed from L. Moroney's Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning

Supervised vs Unsupervised

Supervised The algorithms build a mathematical model of a set of data including...

- inputs
- desired outputs

Unsupervised The algorithms take a set of data that contains...

- only inputs
- ...and find structure in the data

https://en.wikipedia.org/wiki/Machine_learning

Naïve Bayes'

Naïve Bayes'

1. Considered the first machine learning approach
2. Introduced in the IR community by Maron (1961)
3. It is a **supervised** model
4. It applies Bayes' theorem with strong (naïve) independence assumptions between the features:
 - they are independent
 - they contribute "the same"

Naïve Bayes'

A conditional probability model

Given an instance vector

$$\mathbf{x} = (x_1, \dots, x_n) \quad (1)$$

representing n **independent** features $x_1, x_2, x_3, \dots, x_{n-2}, x_{n-1}, x_n$

n could be $|V|$ (the size of the vocabulary)³

The model assigns to instance $\mathbf{x}$ the probability

$$p(C_k | \mathbf{x}) = p(C_k | x_1, \dots, x_n) \quad (2)$$

for each of the k possible outcomes C_k

where $C_k = \{c_1, \dots, c_k\}$

From https://en.wikipedia.org/wiki/Naive_Bayes_classifier

³ $|x|$ is the cardinality of x

Naïve Bayes'

The rest of Naïve Bayes' will be covered in session 5

References

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